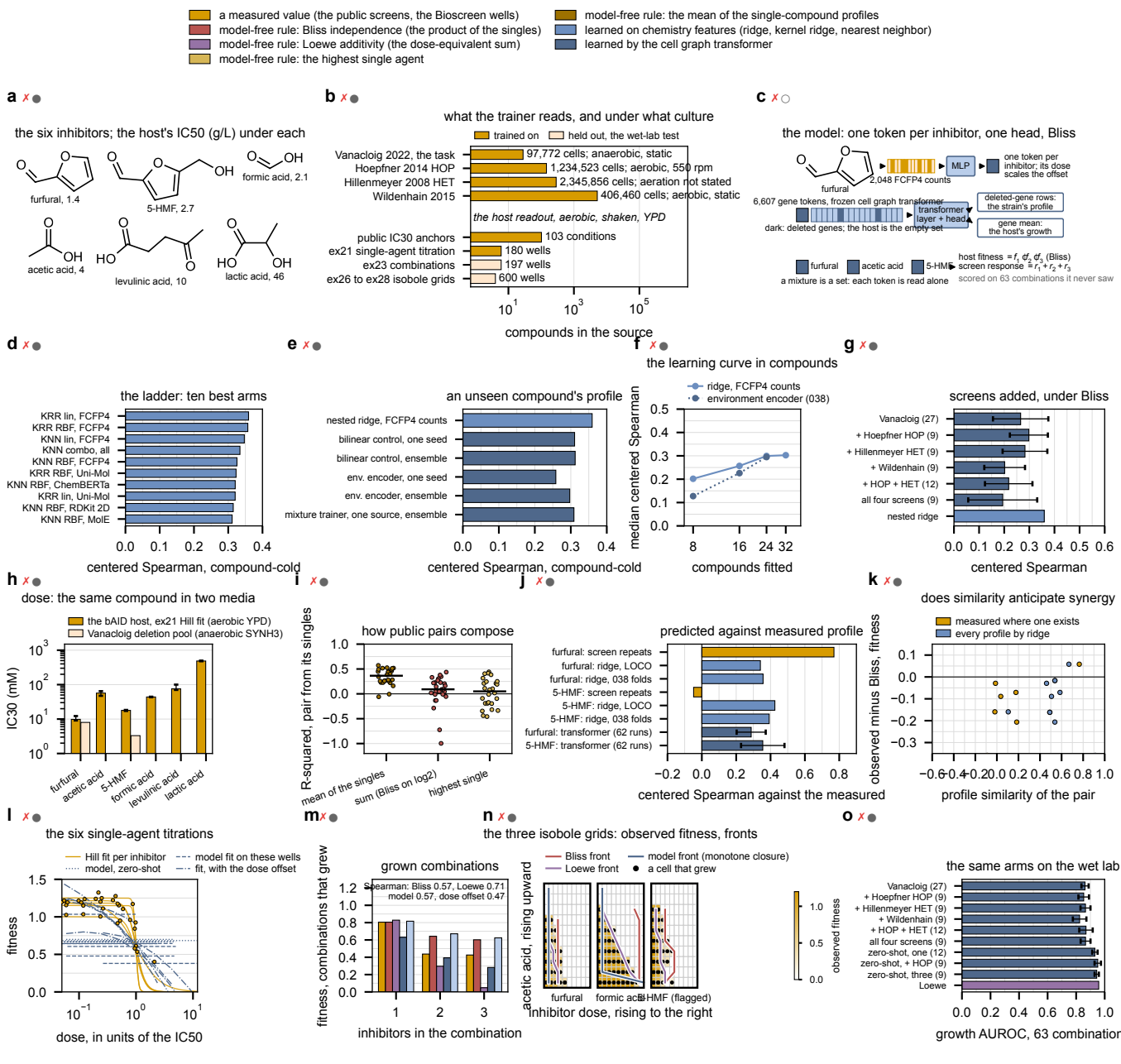

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the six inhibitors; the host's IC50 (g/L) under each
furfural, 1.4
5-HMF, 2.7
formic acid, 2.1
acetic acid, 4
levulinic acid, 10
lactic acid, 46
b
what the trainer reads, and under what culture
trained on
held out, the wet-lab test
Vanacloig 2022, the task
Hoepfner 2014 HOP
Hillenmeyer 2008 HET
Wildenhain 2015
97,772 cells; anaerobic, static
1,234,523 cells; aerobic, 550 rpm
2,345,856 cells; aeration not stated
406,460 cells; aerobic, static
the host readout, aerobic, shaken, YPD
public IC30 anchors
ex21 single-agent titration
ex23 combinations
ex26 to ex28 isobole grids
103 conditions
180 wells
197 wells
600 wells
compounds in the source
c
the model: one token per inhibitor, one head, Bliss
furfural
2,048 FCFP4 counts
MLP
one token per inhibitor; its dose scales the offset
6,607 gene tokens, frozen cell graph transformer
transformer layer + head
deleted-gene rows: the strain's profile
gene mean: the host's growth
dark: deleted genes; the host is the empty set
furfural
acetic acid
5-HMF
a mixture is a set: each token is read alone
host fitness = $f_1 \phi_2 \phi_3$ (Bliss)
screen response = $r_1 + r_2 + r_3$
scored on 63 combinations it never saw
d
the ladder: ten best arms
KRR lin, FCFP4
KRR RBF, FCFP4
KNN lin, FCFP4
KNN combo, all
KNN RBF, FCFP4
KRR RBF, Uni-Mol
KNN RBF, ChemBERTa
KRR lin, Uni-Mol
KNN RBF, RDKit 2D
KNN RBF, MolE
centered Spearman, compound-cold
e
an unseen compound's profile
nested ridge, FCFP4 counts
bilinear control, one seed
bilinear control, ensemble
env. encoder, one seed
env. encoder, ensemble
mixture trainer, one source, ensemble
centered Spearman, compound-cold
f
the learning curve in compounds
ridge, FCFP4 counts
environment encoder (038)
median centered Spearman
compounds fitted
g
screens added, under Bliss
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Wildenhain (9)
+ HOP + HET (12)
all four screens (9)
nested ridge
centered Spearman
h
dose: the same compound in two media
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
i
how public pairs compose
R-squared, pair from its singles
mean of the singles
sum (Bliss on log2)
highest single
j
predicted against measured profile
furfural: screen repeats
furfural: ridge, LOCO
furfural: ridge, 038 folds
5-HMF: screen repeats
5-HMF: ridge, LOCO
5-HMF: ridge, 038 folds
furfural: transformer (62 runs)
5-HMF: transformer (62 runs)
centered Spearman against the measured
k
does similarity anticipate synergy
measured where one exists
every profile by ridge
observed minus Bliss, fitness
profile similarity of the pair
l
the six single-agent titrations
Hill fit per inhibitor
model, zero-shot
model fit on these wells
fit, with the dose offset
fitness
dose, in units of the IC50
m
grown combinations
Spearman: Bliss 0.57, Loewe 0.71
model 0.57, dose offset 0.47
fitness, combinations that grew
inhibitors in the combination
n
the three isobole grids: observed fitness, fronts
Bliss front
Loewe front
model front (monotone closure)
a cell that grew
acetic acid, rising upward
furfural
formic acid
5-HMF (flagged)
inhibitor dose, rising to the right
observed fitness
o
the same arms on the wet lab
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Wildenhain (9)
+ HOP + HET (12)
all four screens (9)
zero-shot, one (12)
zero-shot, + HOP (9)
zero-shot, three (9)
Loewe
growth AUROC, 63 combinations